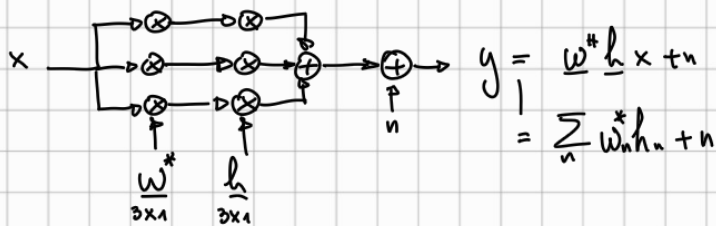


MRT

1 user, 3 antennas



power constraint: $\sum_n |w_n|^2 = \|\underline{w}\|^2 = 1$

SNR: $\gamma = \frac{E_s}{\sigma_n^2} \cdot \left| \sum_n \underline{w}_n^* \underline{h}_n \right|^2 = \frac{E_s}{\sigma_n^2} \cdot \left| \underline{w}^H \underline{h} \right|^2$

→ optimization with Lagrange method on the constraint $\|\underline{w}\|^2 - 1 = 0$

$$\mathcal{L} = \frac{E_s}{\sigma_n^2} \left(\sum_n \underline{w}_n^* \underline{h}_n \right)^2 + \lambda \left(\sum_n \underline{w}_n^2 - 1 \right)$$

$$= \frac{E_s}{\sigma_n^2} \left| \underline{w}^H \underline{h} \right|^2 + \lambda (\|\underline{w}\|^2 - 1)$$

$$\rightarrow \frac{\partial \mathcal{L}}{\partial \underline{w}_m^*} = \frac{E_s}{\sigma_n^2} \left(\sum_n \underline{w}_n^* \underline{h}_n \right)^* \underline{h}_m + \lambda \underline{w}_m = 0$$

$$\underline{w}_m = -\frac{1}{\lambda} \frac{E_s}{\sigma_n^2} \left(\sum_n \underline{w}_n^* \underline{h}_n \right)^* \underline{h}_m$$

$$\rightarrow \frac{\partial \mathcal{L}}{\partial \underline{w}^*} = \frac{E_s}{\sigma_n^2} \cdot \nabla (\underline{w}^H \underline{h}) \underline{h} + \lambda \underline{w} = 0$$

$$\underline{w} = -\frac{1}{\lambda} \frac{E_s}{\sigma_n^2} (\underline{w}^H \underline{h}) \underline{h}$$

$$\Rightarrow \underline{w}_m = c \underline{h}_m \quad \text{with } \sum_n |w_n|^2 = 1 \quad \rightarrow c^2 \sum |h_n|^2 = 1 \quad \rightarrow c = \frac{1}{\sqrt{\sum |h_n|^2}}$$

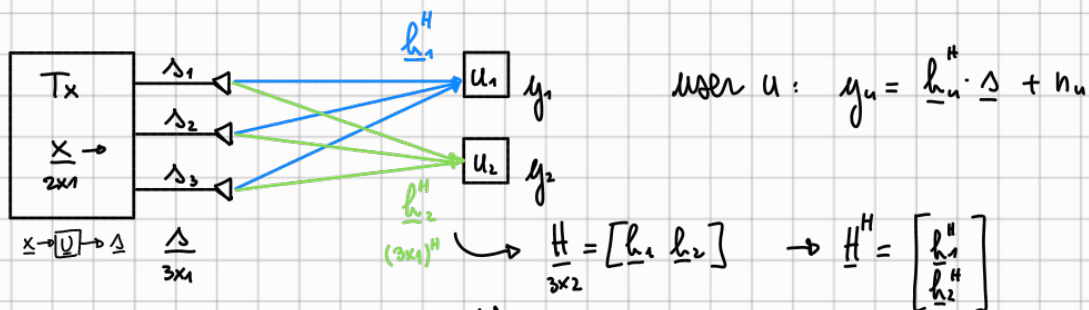
$$\underline{w} = c \underline{h} \quad \|\underline{w}\|^2 = 1 \quad \rightarrow c^2 \|\underline{h}\|^2 = 1 \quad \rightarrow c = \frac{1}{\|\underline{h}\|}$$

$$\rightarrow \underline{w}^H \underline{h} = \frac{\underline{h}^H \underline{h}}{\|\underline{h}\|} = \frac{\|\underline{h}\|^2}{\|\underline{h}\|} = \|\underline{h}\|$$

$$\Rightarrow \text{SNR: } \gamma = \frac{E_s}{\sigma_n^2} (\underline{w}^H \underline{h}) = \frac{E_s}{\sigma_n^2} \|\underline{h}\|^2$$

ZF

2 users, 3 antennas



$$\hookrightarrow \underline{\Delta} = \sum_u \underline{u}_u x_u = \underline{U} \underline{x}$$

at user u : $y_u = \underline{h}_u^H \underline{\Delta} + n_u = \underline{h}_u^H \underline{U} \underline{x} + n_u = \underline{h}_u^H \underline{u}_u x_u + \underbrace{\underline{h}_u^H \cdot \sum_{i \neq u} \underline{u}_i x_i}_{\text{inter-user interf.} \rightarrow 0} + n_u$

$$\underline{y} = \underline{H}^H \underline{U} \underline{x} + \underline{n}$$

$$\hookrightarrow \underline{H}^H \underline{U} = \begin{bmatrix} \underline{h}_1^H \\ \underline{h}_2^H \end{bmatrix} \begin{bmatrix} \underline{u}_1 & \underline{u}_2 \end{bmatrix} = \begin{bmatrix} \underline{h}_1^H \underline{u}_1 & \underline{h}_1^H \underline{u}_2 \\ \underline{h}_2^H \underline{u}_1 & \underline{h}_2^H \underline{u}_2 \end{bmatrix}$$

$$\hookrightarrow \underline{h}_k^H \underline{u}_i = 0 \text{ for } k \neq i$$

$\underline{u}_u = \frac{\underline{f}_u}{\|\underline{f}_u\|}$ with $\underline{f}_u = \underline{H}(\underline{H}^H \underline{H})^{-1} \underline{e}_u \rightarrow \underline{F} = \underline{H}(\underline{H}^H \underline{H})^{-1}$

$$\hookrightarrow \underline{h}_u^H \underline{u}_u = \underline{h}_u^H \underline{f}_u \cdot \frac{1}{\|\underline{f}_u\|} \longrightarrow \underline{h}_u^H \underline{f}_i = 0 \text{ if } u \neq i \text{ to remove inter-user interf.}$$

$\hookrightarrow$ proof of $\underline{h}_u^H \underline{f}_i = 0$ and $\underline{h}_u^H \underline{u}_i = 0$

$$\begin{aligned} \underline{h}_u^H \underline{f}_i &= \\ &= \underline{h}_u^H \underline{H}(\underline{H}^H \underline{H})^{-1} \underline{e}_i = \\ &= \underline{h}_u^H [\underline{h}_1 \ \underline{h}_2] \left(\begin{bmatrix} \underline{h}_1^H \\ \underline{h}_2^H \end{bmatrix} \begin{bmatrix} \underline{h}_1 & \underline{h}_2 \end{bmatrix} \right)^{-1} \underline{e}_i \end{aligned}$$

$$= \underline{h}_u^H [\underline{h}_1 \ \underline{h}_2] \begin{bmatrix} \underline{h}_1^H \underline{h}_1 & \underline{h}_1^H \underline{h}_2 \\ \underline{h}_2^H \underline{h}_1 & \underline{h}_2^H \underline{h}_2 \end{bmatrix}^{-1} \underline{e}_i$$

$$= \underline{h}_u^H [\underline{h}_1 \ \underline{h}_2] \begin{bmatrix} \|\underline{h}_1\|^2 & \underline{h}_1^H \underline{h}_2 \\ \underline{h}_2^H \underline{h}_1 & \|\underline{h}_2\|^2 \end{bmatrix}^{-1} \underline{e}_i$$

$$= \underline{h}_1^H [\underline{h}_1 \ \underline{h}_2] \cdot \frac{1}{|\underline{H}^H \underline{H}|} \begin{bmatrix} \|\underline{h}_2\|^2 & -\underline{h}_1^H \underline{h}_2 \\ -\underline{h}_2^H \underline{h}_1 & \|\underline{h}_1\|^2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\hookrightarrow u=1, i=2$$

$$\begin{aligned}
&= \frac{1}{|\underline{H}^H \underline{H}|} \underline{h}_1^H [\underline{h}_1 \ \underline{h}_2] \begin{bmatrix} -\underline{h}_1^H \underline{h}_2 \\ \|\underline{h}_1\|^2 \end{bmatrix} \\
&= \frac{1}{|\underline{H}^H \underline{H}|} \underline{h}_1^H [-\underline{h}_1 \underline{h}_1^H \underline{h}_2 + \|\underline{h}_1\|^2 \underline{h}_2] \\
&= \frac{1}{|\underline{H}^H \underline{H}|} \underbrace{(-\|\underline{h}_1\|^2 \underline{h}_1^H \underline{h}_2 + \|\underline{h}_1\|^2 \underline{h}_1^H \underline{h}_2)}_{=0} = 0
\end{aligned}$$

$$\begin{aligned}
\hookrightarrow \underline{h}_u \underline{f}_u &= \\
&= \underline{h}_u^H \underline{H} (\underline{H}^H \underline{H})^{-1} \underline{e}_u = \\
&= \underline{h}_u^H [\underline{h}_1 \ \underline{h}_2] \left(\begin{bmatrix} \underline{h}_1^H \\ \underline{h}_2^H \end{bmatrix} [\underline{h}_1 \ \underline{h}_2] \right)^{-1} \underline{e}_u \\
&= \underline{h}_u^H [\underline{h}_1 \ \underline{h}_2] \begin{bmatrix} \underline{h}_1^H \underline{h}_1 & \underline{h}_1^H \underline{h}_2 \\ \underline{h}_2^H \underline{h}_1 & \underline{h}_2^H \underline{h}_2 \end{bmatrix}^{-1} \underline{e}_u \\
&= \underline{h}_u^H [\underline{h}_1 \ \underline{h}_2] \begin{bmatrix} \|\underline{h}_1\|^2 & \underline{h}_1^H \underline{h}_2 \\ \underline{h}_2^H \underline{h}_1 & \|\underline{h}_2\|^2 \end{bmatrix}^{-1} \underline{e}_u \quad \rightarrow u=i=1 \\
&= \underline{h}_1^H [\underline{h}_1 \ \underline{h}_2] \cdot \frac{1}{|\underline{H}^H \underline{H}|} \begin{bmatrix} \|\underline{h}_2\|^2 & -\underline{h}_1^H \underline{h}_2 \\ -\underline{h}_2^H \underline{h}_1 & \|\underline{h}_1\|^2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\
&= \underline{h}_1^H [\underline{h}_1 \ \underline{h}_2] \cdot \frac{1}{|\underline{H}^H \underline{H}|} \begin{bmatrix} \|\underline{h}_2\|^2 \\ -\underline{h}_2^H \underline{h}_1 \end{bmatrix} \\
&= \frac{1}{|\underline{H}^H \underline{H}|} \underline{h}_1^H [\underline{h}_1 \|\underline{h}_2\|^2 - \underline{h}_2 \underline{h}_2^H \underline{h}_1] \\
&= \frac{\|\underline{h}_2\|^2 \|\underline{h}_2\| - \underline{h}_1^H \underline{h}_2 \underline{h}_2^H \underline{h}_1}{|\underline{H}^H \underline{H}|}
\end{aligned}$$

• SINR for user u :

$$\text{SINR}_u = \frac{|\underline{h}_u^H \underline{u}_u|^2 E_u}{\sum_{i \neq u} |\underline{h}_u^H \underline{u}_i|^2 E_i + \sigma_n^2} = \frac{|\underline{h}_u^H \underline{f}_u \cdot \frac{1}{\|\underline{f}_u\|^2}|^2 E_u}{\sigma_n^2} = \frac{E_u}{\sigma_n^2} \frac{|\underline{h}_u^H \underline{f}_u|^2}{\|\underline{f}_u\|^2}$$

• rate for user u : $R_u = \log_2 (1 + \text{SINR}_u)$

$$= \log_2 \left(1 + \frac{E_u}{\sigma_n^2} \frac{|\underline{h}_u^H \underline{f}_u|^2}{\|\underline{f}_u\|^2} \right)$$

• sum rate: $R = \sum_u \log_2 \left(1 + \frac{E_u}{\sigma_n^2} \frac{|\underline{h}_u^H \underline{f}_u|^2}{\|\underline{f}_u\|^2} \right)$

$$= \sum_u \log_2 \left(1 + \frac{E_u}{\sigma_n^2} \frac{|\underline{h}_u \underline{H} (\underline{H}^H \underline{H})^{-1} \underline{e}_u|^2}{\|\underline{H} (\underline{H}^H \underline{H})^{-1} \underline{e}_u\|^2} \right)$$

